

Joint Training with Machine-Generated Text Detection Can Distort Sarcasm Detection: A Bilingual Diagnosis of Class-Conditional Negative Transfer

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Abstract

Sarcasm is non-literal language whose intended meaning may diverge from apparent sentiment. As a characteristically human form of expression grounded in implicit intent and context, it is difficult to detect automatically. LLM-generated examples could improve detection by augmenting training data. However, this strategy requires LLMs to produce text that both conveys sarcasm and resembles human writing, so sarcasm and authorship must be evaluated together. Although sarcasm detection and machine-generated text detection (MGTD) are usually studied separately, their interaction in shared representations remains insufficiently understood. MGTD must encode source cues, whereas sarcasm detection should reflect intent rather than source. To test whether this interaction is language-specific, we provide the first bilingual diagnosis, to our knowledge, of joint MGTD–sarcasm training in Chinese and English. We harmonize bilingual resources, generate topic-matched AI text, and build DualTest with source and sarcasm labels, defining AI labels by requested style. Compared with their single-task counterparts, jointly trained models are markedly more likely to misclassify AI-generated texts prompted to be non-sarcastic as sarcastic. Human-authorship detector scores and LLM human-likeness ratings also rank outputs differently. These findings motivate subgroup evaluation and caution against detector-only assessment.

1 Introduction

A sarcastic sentence conveys an attitude or intent that may oppose its literal meaning. As a characteristically human form of expression, sarcasm relies on implicit contrast, intent, and context, making it difficult to detect automatically. Errors can invert sentiment analysis and disrupt opinion-mining or content-moderation systems (Tay et al., 2018; Vidgen et al., 2019). Annotation outcomes also

vary with elicitation method and annotator perspective (Oprea and Magdy, 2020; Casola et al., 2024), while model behavior is sensitive to domain, style, and language (Jang and Frassinelli, 2024; Casola et al., 2024) (§2.1).

One potential response is to augment training data with LLM-generated examples. This strategy is useful only if generated text both conveys the intended sarcasm and resembles human writing; otherwise, source-specific artifacts may become shortcuts. Sarcasm and authorship must therefore be considered together. Human-authored and AI-generated texts can each be sarcastic or non-sarcastic, yielding all four combinations.

Machine-generated text detection (MGTD) imposes a contrasting representational demand: it must distinguish human-written from AI-generated text and thus favors source-predictive representations (Guo et al., 2023; Wang et al., 2024) (§2.2). These stylistic or distributional cues may overlap with features used for sarcasm detection. The labels do not conflict; the potential conflict lies in how the tasks organize a shared representation. MGTD must encode source, whereas sarcasm detection should reflect intent rather than source. Shared-encoder multi-task learning can suffer task conflict and negative transfer by exploiting information useful for one task but spurious for another (Mueller et al., 2022; Hu et al., 2022b). Separate models avoid this coupling but forgo efficient encoder sharing, motivating a direct test of whether joint training contaminates sarcasm decisions.

Detector scores also matter downstream because they can become generation rewards. ViSP derives PPO rewards from a sarcasm detector (Wang et al., 2026). Yet useful generated sarcasm must satisfy two distinct objectives: conveying sarcastic intent and being perceived as human-like. A dual-head detector may appear to provide corresponding signals through sarcasm and authorship probabilities,

but authorship probability is not itself a measure of human-likeness. We therefore test whether these detector scores track source-blinded LLM ratings of sarcasm and perceived human-likeness (§6).

To test these risks rather than assume MGTD is a beneficial auxiliary task, we compare same-architecture dual- and single-head models on bilingual Chinese–English data and build DualTest for source $\times$ sarcasm subgroup analysis. The evaluated joint-training procedure sharply increases sarcasm false positives on AI text prompted to be non-sarcastic; we call this class-conditional negative transfer. The direction persists when the AI test texts are regenerated with Qwen3.5-9B, while RCNN and interaction projection show lower rates than the plain configuration.

We then examine the shared representation. Source information is highly decodable and correlates with sarcasm decisions. To connect encoded information to actual decisions beyond probe accuracy (Hewitt and Liang, 2019; Elazar et al., 2021), we combine probes, task-head correlations, and representation interventions. These analyses identify a non-random high-variance direction to which the subgroup gap is sensitive, supporting a source-correlated shortcut hypothesis and providing a concrete target for causal analysis (§5.3.2, Appendix E).

Contributions. To our knowledge, we provide the first bilingual diagnostic study of how joint training with MGTD changes sarcasm detection across Chinese and English, together with DualTest for source $\times$ sarcasm analysis. Same-architecture comparisons across models and generators identify class-conditional negative transfer concentrated on AI-generated text prompted to be non-sarcastic; cross-architecture comparisons characterize how its severity varies across the evaluated configurations. Representation analyses support the source-correlated shortcut hypothesis and define concrete targets for causal analysis. Finally, an exploratory detector–judge comparison illustrates a downstream risk of using detector scores to evaluate generated sarcasm and perceived human-likeness.

2 Related Work

2.1 Sarcasm Detection and Multilingual Resources

Feature-engineered methods (Riloff et al., 2013), neural sequence models (Ghosh and Veale, 2016;

Tay et al., 2018), and pretrained language models (Khatri and P, 2020; Du et al., 2022) have each advanced sarcasm detection. The task remains difficult in multilingual and class-imbalanced settings, especially because datasets differ in whether labels reflect intended sarcasm or annotator-perceived sarcasm (Oprea and Magdy, 2020; Abu Farha et al., 2022; Casola et al., 2024). Although scores are not directly comparable across datasets, languages, modalities, or evaluation setups, they illustrate substantial benchmark variability. On English iSarcasmEval (Abu Farha et al., 2022), the top two systems report positive-class F1 of 0.605 and 0.569; across MultiPiCo languages (Casola et al., 2024), zero-shot ChatGPT positive-class F1 ranges from 0.152 to 0.567. On the bilingual multimodal SarcNet benchmark (Yue et al., 2024), a text-only XLM-R baseline reaches 0.734 F1.

RCNNs combine recurrent context with convolutional pooling for text classification (Lai et al., 2015) and have also been evaluated for multilingual sarcasm detection (Yang and Ikeda, 2024); we therefore include an RCNN intermediate configuration in §4.1.

The benchmark results summarized above come from different datasets, so §5.1 provides directly comparable mBERT dual-head baselines on DualTest. To our knowledge, existing sarcasm resources do not provide both sarcasm and human-versus-AI source labels. DualTest supplies both labels in a bilingual Chinese–English setting.

The Chinese and English source resources differ in domain, annotation procedure, and class balance, so bilinguality introduces additional distributional variation. We treat language as a robustness and stratification dimension rather than as the primary transfer objective: the study asks whether the subgroup failure is confined to one language, not whether knowledge transfers across languages. We therefore report language-stratified results in addition to pooled metrics.

2.2 Machine-Generated Text Detection and Authorship Verification

Machine-generated text detection (MGTD) comprises zero-shot model-likelihood methods, such as probability-curvature detectors (Mitchell et al., 2023), and discriminative models trained on paired human–AI corpora (HC3; Guo et al., 2023). Generalization is a central difficulty: M4 (Wang et al., 2024) provides a benchmark across generators, domains, and languages and shows that detectors

struggle on unseen domains or LLMs, while RAID (Dugan et al., 2024) stress-tests detectors under unseen generators, decoding variations, and adversarial attacks. This generalization gap is consistent with reliance on source-predictive regularities that are partly generator- or domain-specific; such cues are relevant here because they may enter a representation shared with sarcasm detection. Bevendorff et al. (2025) distinguish closed-set authorship attribution from open-set authorship verification and argue that realistic LLM detection is better cast as the latter. Our binary MGTD setup is narrower: we use it as a controlled source-classification auxiliary task, not as a measure of text quality.

In our literature search, we found no work that jointly trains sarcasm detection and MGTD with a shared encoder.¹ This gap motivates our study because source information has different roles in the two tasks: it is the target of MGTD but should not by itself determine a sarcasm detector’s prediction. Our study tests whether a shared representation preserves that distinction.

2.3 Multi-Task Negative Transfer, Shortcut Learning, and Shared-Encoder Mitigation

In multi-task learning, auxiliary tasks are not always beneficial: transfer depends on task relatedness and alignment between task data (Zhang et al., 2023; Wu et al., 2020), while gradient conflict is a central optimization challenge that can contribute to negative transfer (Yu et al., 2020; Liu et al., 2021). Mueller et al. (2022) provide a relevant NLP precedent through matched single- and multi-task comparisons. They find similar negative-transfer patterns in canonical and text-to-text systems and note that models may infer task identity from input distributions, a strategy that may not transfer reliably out of distribution. He and Choi (2021) and Li et al. (2023) further analyze functional specialization in multi-task Transformer encoders. Shortcut learning offers a complementary reading: models may exploit features that are predictive in the training distribution yet fail under distribution shift (Geirhos et al., 2020; Ye et al., 2026). In MTL, shared representations can increase reliance on spurious correlations (Hu

et al., 2022b; Chai et al., 2025). Related multi-dataset work shows that adding data can introduce spurious correlations and reduce worst-group performance (Compton et al., 2023).

Shared-encoder mitigation either isolates task-specific parameters (Pfeiffer et al., 2021, 2020; Karimi Mahabadi et al., 2021; Stickland and Murray, 2019) or modifies information flow within shared computation (Wu et al., 2019; Pilault et al., 2021). Our analysis is closer to the second design direction: we compare a no-module baseline with two intermediate-representation variants and test whether degradation changes across source $\times$ sarcasm subgroups (§5.3).

2.4 Misalignment in Detector-Based Evaluation and Rewards

The representational conflict studied above may also matter when detector outputs are reused beyond classification, particularly as evaluation or reward signals for generated text. Automated scorers, including LLM judges, are increasingly used to evaluate generated text (Zheng et al., 2023). Model-based scores are also optimized as rewards, but overoptimization can increase a proxy score while degrading a gold-standard measure (Gao et al., 2023). A detector can achieve useful classification accuracy without its continuous output being calibrated as a measure of style strength or generation quality. Multimodal sarcasm generation may face this risk: ViSP (Wang et al., 2026) uses the DIP sarcasm detector (Wen et al., 2023) as a PPO reward (Schulman et al., 2017). As an exploratory downstream analysis, Section 6 compares detector scores for human authorship with separately obtained, source-blinded LLM ratings of perceived human-likeness.

3 Dataset Construction

Testing for interference requires separate task supervision and evaluation labels for authorship and sarcasm. Here “source” denotes human-versus-AI authorship; “corpus” denotes an input dataset. We combine iSarcasm (Oprea and Magdy, 2020; Abu Farha et al., 2022), CSC (Jang and Frassinelli, 2024; Jang et al., 2023), ToSarcasm (Liang et al., 2022), News Headlines v2 (Misra and Arora, 2023; Misra and Grover, 2021), and the Open Chinese Internet Sarcasm Corpus (Zhu, 2022). None provides both labels. Mapping their annotations to a common binary sarcasm label yields 44,862

¹Searches conducted in May 2026 and updated in July 2026 across Semantic Scholar, arXiv, ACL Anthology, and Google Scholar, using keyword combinations including “sarcasm detection,” “machine-generated text detection,” “AI-generated text detection,” “multi-task learning,” and “joint detection.”

pooled instances (approximately 87% English), from which we construct three training sets and DualTest (Appendix A).

Sarcasm classification training set, SarcTrain (10,000, all human). Each language $\times$ label bucket contains 2,500 instances. Three buckets are downsampled; the Chinese sarcastic bucket starts from 2,396 records and adds 104 oversampled records. The resulting set is organized into five folds.

MGTD training set, AuthTrain (19,995). AuthTrain draws 9,999 approximately language-balanced human seeds from SarcTrain (5,000 English / 4,999 Chinese) after one Chinese seed is excluded during seed-level generation quality control. The Ollama model tagged deepseek-r1:8b (DeepSeek-AI, 2025; Ollama, 2025) produces 9,996 retained AI counterparts; three selected human seeds remain unmatched, giving 99.97% pairing coverage. Appendix A details the exclusions.

Four language $\times$ style templates follow the balanced seed labels. Authorship and requested-style labels form two separately controlled, approximately balanced factors, enabling source $\times$ sarcasm subgroup analysis (Appendix A).

Generation training set, GenTrain (2,428; used in §6). From 5,257 eligible positive-sarcasm records in iSarcasm and ToSarcasm, a 40% sample selects 2,103 records (1,214 English; 889 Chinese). Chinese oversampling adds 325 records, yielding 1,214 per language. The resulting set contains 374 rewriting, 839 response, and 1,215 comment instances and serves as the instruction-tuning data for supervised fine-tuning (SFT; Appendix A).

Diagnostic test set, DualTest (4,664). The human subset combines official iSarcasm (1,400 English) and ToSarcasm (972 Chinese) test sets. Generation quality control retains AI counterparts for 2,292 of 2,372 texts (96.6% coverage). Human texts retain their annotations; for AI texts, the sarcasm label records the prompted condition rather than an independently annotated judgment of perceived sarcasm.

The four source $\times$ sarcasm groups contain 823 human-sarcastic, 1,549 human-non-sarcastic, 779 AI-sarcastic, and 1,513 AI-non-sarcastic instances. Appendix A gives the language $\times$ source $\times$ sarcasm distribution.

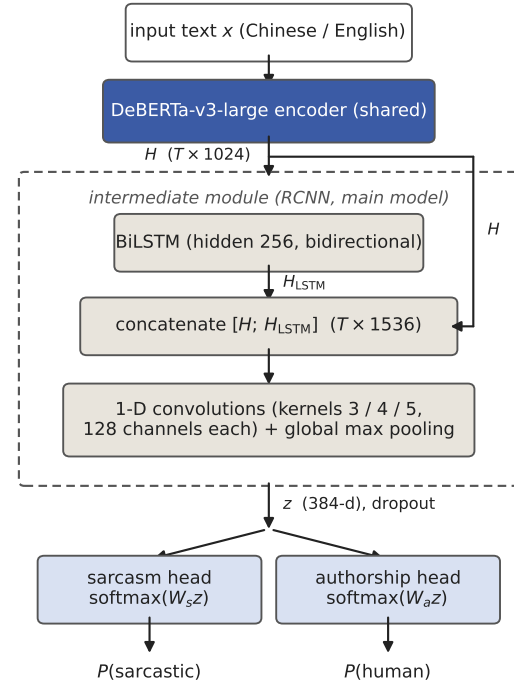


Figure 1: RCNN dual-head configuration: a shared DeBERTa-v3-large encoder, an RCNN intermediate module (dashed), and two task-specific heads. The comparisons in §5 remove this component in the plain configuration or replace it with task-branch interaction projection, while retaining the same encoder backbone and task definitions (Appendix B).

4 Dual-Head Detection Model

4.1 Tasks and Architecture

Given a Chinese or English input text x , the model produces two separate, non-mutually-exclusive binary predictions: whether x is labeled sarcastic and whether its authorship source is labeled human or AI. Any combination of the two labels is valid. The tasks share a single DeBERTa-v3-large encoder (He et al., 2023), allowing us to test how incorporating MGTD into joint training changes the representation read by the sarcasm head. Encoder outputs are aggregated by an intermediate module and then passed to two task-specific classification heads; Figure 1 shows the RCNN configuration.

The RCNN configuration uses BiLSTM + CNN as its intermediate module. A single-layer bidirectional LSTM with hidden size 256 per direction produces a 512-dimensional output at each token position. We concatenate it with the 1,024-dimensional encoder output, then apply three groups of one-dimensional convolutions with ker-

nel widths 3/4/5 and 128 channels each. Global max pooling produces a 384-dimensional vector z . After dropout, z is passed to the two heads, yielding a DeBERTa-based dual-head RCNN configuration.

To test the intermediate layer’s role, §5 compares two variants. The plain model sends [CLS] directly to both heads. The interaction-projection model forms two learned 256-dimensional projections, h_s and h_a , and updates them as $h'_s = h_s + P_{a \rightarrow s}(h_a)$ and $h'_a = h_a + P_{s \rightarrow a}(h_s)$. Because each branch supplies a single key–value vector, its attention weight is identically one; the module therefore reduces to residual cross-branch affine projections, where $P_{a \rightarrow s}$ and $P_{s \rightarrow a}$ each compose the attention value and output projections. The configurations retain the same encoder and task definitions while testing distinct intermediate-representation designs (Appendix B).

4.2 Training Protocol

The training protocol supports controlled dual-single comparisons in the plain and RCNN configurations. Within each comparison, the models use the same encoder architecture, intermediate configuration, sarcasm data and fold assignments, and sarcasm-specific hyperparameters. The dual model additionally receives AuthTrain batches, updates shared parameters from both tasks, follows the mixed-task schedule, and uses joint early stopping; the comparison therefore evaluates the complete joint-training procedure rather than any single component.

Training uses a mixed loader that supplies task-homogeneous batches according to a 2:1 MGTD:sarcasm schedule, matching the approximate data-size ratio between AuthTrain and SarcTrain. Let θ_{sh} denote parameters updated by both tasks, let $z_s(x)$ and $z_a(x)$ be the representations read by the sarcasm and MGTD heads, and let W_s and W_a denote the two classification heads. The two representations coincide in the plain and RCNN configurations; in interaction projection they are the task-specific h'_s and h'_a defined above. At step t , the loss is computed only for the task corresponding to the current batch B_t :

$$\mathcal{L}(B_t) = \begin{cases} \frac{1}{|B_t|} \sum_{(x,y) \in B_t} \text{CE}_w(W_s z_s(x), y), & B_t \subset \text{SarcTrain}, \\ \frac{1}{|B_t|} \sum_{(x,y) \in B_t} \text{CE}(W_a z_a(x), y), & B_t \subset \text{AuthTrain}, \end{cases} \quad (1)$$

where CE_w is a pre-specified cost-sensitive cross-entropy for sarcasm (weights are provided in Appendix B). With SarcTrain class-balanced, the weighted loss assigns a greater penalty to errors on sarcastic examples. Softmax is applied to logits only when probabilities are required. The gradient flow follows from the equation: on sarcasm batches, $\partial \mathcal{L} / \partial W_a = 0$ (symmetrically, $\partial \mathcal{L} / \partial W_s = 0$ on MGTD batches), whereas θ_{sh} receives gradients from both types of batches. Unless otherwise stated, binary predictions use a fixed probability threshold of 0.5 rather than a threshold tuned on DualTest.

The encoder uses a lower learning rate than newly added modules. The sarcasm-head rate is fixed across configurations; MGTD-head rates are reported in Appendix B.

Model selection accounts for both tasks. At epoch e , the early-stopping signal is the equally weighted sum of the two validation losses, each normalized by its first-epoch value:

$$\mathcal{L}_{\text{stop}}^{(e)} = \frac{1}{2} \frac{\mathcal{L}_s^{(e)}}{\mathcal{L}_s^{(1)}} + \frac{1}{2} \frac{\mathcal{L}_a^{(e)}}{\mathcal{L}_a^{(1)}}, \quad (2)$$

where $\mathcal{L}_s^{(e)}$ and $\mathcal{L}_a^{(e)}$ are the epoch- e validation losses for the sarcasm and MGTD tasks on their respective validation sets. Normalization reduces the influence of differences in task-specific loss scales and learning dynamics.

Five runs are defined by the SarcTrain validation folds: four folds train and one validates sarcasm, while AuthTrain uses a fixed 90/10 split. Each run reloads the pretrained encoder and randomly initializes new modules and heads. Configurations share fold assignments and nominal seeds, enabling split-matched comparisons but not identical initialization.

DualTest enters no training or validation split for the sarcasm or MGTD models. Five-run standard deviations combine fold and initialization variation. Wilson intervals are conditional on a fixed final model and cover test-instance, not training or generator, uncertainty (Table 3; Appendix C.7).

Each subgroup configuration has a final model trained on all applicable data with a fixed epoch count determined from the five validation runs, an independent seed, and no early stopping. Their weights and training duration are not selected using DualTest, and they are not used in five-run aggregates; §6 uses the final RCNN dual-head model (Appendix B).

Model	Sarc. F1	MGTD F1
mBERT dual-head	0.4945 $\pm$.036	0.8743 $\pm$.006
XLM-RoBERTa-base dual-head	0.4603 $\pm$.057	0.8321 $\pm$.025
DeBERTa-v3-base dual-head	0.5646 $\pm$.056	0.8952 $\pm$.004
DeBERTa-v3-large plain dual-head	0.6126 $\pm$.031	0.9221 $\pm$.010
+ interaction projection	0.6076 $\pm$.020	0.9126 $\pm$.007
+ RCNN	0.6309 $\pm$.020	0.9301 $\pm$.005

Table 1: Descriptive dual-head results across backbones and architectural variants (DualTest; five-run mean $\pm$ standard deviation; bold = highest mean).

5 Experiments and Diagnosis

With the sarcasm and MGTD models fixed, we use DualTest first to compare encoder backbones and intermediate configurations, then to measure the change between the evaluated joint- and single-task procedures, and finally to partition predictions into source $\times$ sarcasm groups. Aggregate results are five-run means $\pm$ standard deviations; fixed final models support paired instance-level subgroup comparisons and Wilson intervals.

5.1 Backbone Comparison

We compare dual-head mBERT (Devlin et al., 2019), XLM-RoBERTa-base (Conneau et al., 2020), DeBERTa-v3-base, and DeBERTa-v3-large (He et al., 2023) under one evaluation protocol, adding the two intermediate variants on the large encoder. Positive-class F1 uses the sarcastic class for sarcasm and the human class for MGTD on full DualTest.

Table 1 shows higher mean F1 for the evaluated DeBERTa configurations than for mBERT and XLM-R. RCNN has the highest means, reaching 0.6309 sarcasm F1 and 0.9301 MGTD F1. Across the five runs, the two intermediate variants also have 34–47% lower observed standard deviations than the plain large model, providing configuration-level evidence of lower run-to-run variation. The comparison evaluates complete configurations that jointly differ in backbone identity, scale, and intermediate representation. For the separately trained final RCNN model, sarcasm F1 is 0.819 in Chinese and 0.429 in English; prevalence, genre, annotation, and language all contribute to this difference, so it should not be read as a pure language effect. Appendix C provides macro-averaged, human-subset, and language-stratified results.

Arch.	Task	Dual	Single	Δ
Plain	Sarc.	0.6126 $\pm$.031	0.6734 $\pm$.010	−6.08
Plain	MGTD	0.9221 $\pm$.010	0.9054 $\pm$.027	+1.67
RCNN	Sarc.	0.6309 $\pm$.020	0.6751 $\pm$.022	−4.43
RCNN	MGTD	0.9301 $\pm$.005	0.9253 $\pm$.006	+0.49

Table 2: Dual-head vs. single-head models (mean $\pm$ standard deviation over five fold-based runs; Δ = dual − single, in percentage points).

5.2 Dual-Head versus Single-Head Comparison

Within each architecture, dual- and single-head models share the encoder, intermediate structure, sarcasm data, fold assignments, and sarcasm-specific hyperparameters; dual runs additionally receive AuthTrain batches and MGTD updates. Single-head controls traverse the full sarcasm fold each epoch, whereas dual-head models draw sarcasm batches under the 2:1 schedule (§4.2).

Joint training lowers mean sarcasm F1 by 6.08 pp for the plain model and 4.43 pp for RCNN, whereas mean MGTD F1 changes by only +1.67 and +0.49 pp. In all ten split-matched comparisons—five per architecture—sarcasm F1 is lower for the jointly trained model, with differences from −0.8 to −8.6 pp. The consistent degradation is therefore in sarcasm detection rather than in both tasks. We next identify which source and sarcasm groups account for it. RCNN shows a smaller dual–single gap, motivating capacity- and compute-matched tests of which RCNN components contribute to this difference.

5.3 Source $\times$ Sarcasm Subgroup Analysis

An overall F1 score can hide a failure concentrated in one type of text. We therefore divide DualTest into four groups defined by source and sarcasm label. Human labels are dataset annotations, whereas AI labels record the prompted condition. Because each group contains only one reference class, we report recall for sarcastic texts and false-positive rate for non-sarcastic texts.

5.3.1 Three Architectures and Single-Head Controls

The three dual-head variants show whether the subgroup error changes with the intermediate representation. Plain and RCNN single-head models provide same-architecture controls. Within each pair, both models use the same all-human sarcasm data; only the dual-head model additionally un-

dergoes MGTD training. Their difference therefore measures the complete joint-training procedure. Table 3 reports sarcasm predictions; corresponding MGTD results are in Appendix C.

The plain dual-head model predicts sarcasm for 86.3% of AI $\times$ non-sarcastic instances. The observed rates are 40.9% for interaction projection and 34.1% for RCNN, while both same-architecture single-head controls yield 19.3%. Thus, the observed severity is more than 45 pp lower in the two evaluated intermediate configurations than in the plain dual-head configuration.

Among the same AI non-sarcastic texts, 1,021 are false positives only for the plain dual-head model, whereas 7 are false positives only for its single-head control ($p < .001$). The control receives the same all-human sarcasm supervision but not the joint MGTD procedure, tying the large increase to the evaluated joint-training procedure. RCNN shows the same direction with a smaller gap. The matched contrasts are provided by the plain and RCNN models, while interaction projection adds complementary configuration-level evidence.

No analogous extreme subgroup error appears in the MGTD head: its false-positive rate is 6.1% for plain and 6.6% for RCNN. Within AI-generated texts, sarcasm AUC remains 0.82–0.88 across models, so the relative ranking of sarcastic and non-sarcastic AI texts is largely preserved. Joint training instead shifts predicted sarcasm probabilities upward, causing many texts prompted to be non-sarcastic to cross the fixed 0.5 threshold. The observed failure is therefore an AI-associated score increase on this subgroup, not a complete loss of within-AI ranking ability.

The matched plain comparison makes the concentration especially clear: its non-sarcastic false-positive rate changes by 4.2 pp on human text but by 67.0 pp on AI-generated text. The aggregate F1 decrease is therefore driven by a sharply localized source $\times$ class effect rather than a uniform loss across DualTest.

5.3.2 Mechanism Hypothesis and Representation-Level Diagnostics

We next ask whether source information in the shared representation is connected to the higher sarcasm scores. A linear probe predicts source from both dual-head (AUC 0.986) and same-architecture single-head (0.974) representations. The comparable single-head AUC establishes

source decodability as a background property shared by both models. In the dual-head model, texts with a higher MGTD-head AI logit also tend to have a higher sarcasm logit in the target subgroup ($\rho = 0.519$). After covariance normalization, the source directions estimated from the MGTD head and the probe are also moderately aligned with the sarcasm direction. Together, the logit correlation and directional alignment provide correlational evidence linking source-related structure to the affected sarcasm decision.

We then modify the frozen representations and recompute predictions. LEACE (Belrose et al., 2023), which reduces linearly decodable source information, lowers source-probe AUC from 0.982 to 0.589. It also changes the AI/human non-sarcastic false-positive rates from 84.8%/32.8% to 68.7%/49.5%. The nearly unchanged unweighted mean (58.8% to 59.1%) shows that LEACE specifically reallocates errors between source groups, revealing that the affected directions shape source-conditioned error allocation.

A key control removes a high-variance direction constructed to be orthogonal to the estimated linear source direction. This changes the AI/human false-positive rates to 66.7%/54.1% even though measured source decodability remains 0.982; ten sampled random directions show no comparable change. Together, these controls show prediction sensitivity to a non-random high-variance direction. The source-correlated shortcut hypothesis therefore provides a focused mechanism for further analysis, with the semantic content of the effective direction as the next causal target (Appendix E).

5.3.3 Cross-Generator Replication

We keep the trained detectors fixed and compare the DualTest AI texts generated by deepseek-r1:8b with replacements generated by Qwen3.5-9B (Ollama tag qwen3.5:9b) (Qwen Team, 2026). After quality control, the non-sarcastic subsets generated by deepseek-r1:8b and Qwen3.5-9B contain 1,513 and 1,550 outputs. The plain dual-head false-positive rates are 86.3% and 91.3%, whereas its single-head control gives 19.3% and 7.9%; the dual-single gap therefore changes from 67.0 to 83.4 pp. On the Qwen3.5-generated subset, RCNN and interaction projection remain much lower than the plain dual model at 22.1% and 29.5%. The same gap direction and configuration ordering appear

Model	H sarc. rec.	H non-sarc. FP	AI sarc. rec.	AI non-sarc. FP
Plain dual-head	67.1%	32.9%	98.2%	86.3% [84.5, 88.0]
Interaction-projection dual-head	71.9%	37.6%	94.4%	40.9% [38.5, 43.4]
RCNN dual-head	80.8%	40.5%	89.3%	34.1% [31.8, 36.5]
Plain single-head (sarcasm only)	67.8%	28.7%	76.6%	19.3% [17.4, 21.4]
RCNN single-head (sarcasm only)	66.2%	28.5%	79.1%	19.3% [17.4, 21.4]

Table 3: Five sarcasm classifiers on four source $\times$ sarcasm quadrants (recall / false-positive rate, FP). Human labels are annotations; AI labels record prompted conditions. Brackets give model-conditional Wilson 95% intervals; full intervals and paired tests are in [Appendix C.7](#).

with Qwen3.5, although the magnitudes differ (Appendix C).

6 Exploratory Comparison of Detector Scores and LLM-Judge Ratings

As a downstream implication, we test whether source-classification probability tracks perceived human-likeness and whether sarcasm probability tracks LLM ratings. Human-authorship probability and perceived human-likeness are distinct constructs; their comparison directly tests whether a classifier score can function as a generation-evaluation proxy. We compare 600 outputs from Qwen3-8B (Yang et al., 2025) before and after QLoRA (Detrmers et al., 2023) fine-tuning on GenTrain. The RCNN detector and three LLM judges—DeepSeek-V3 (DeepSeek-AI, 2024), Gemini-2.5-Flash-Lite (Google DeepMind, 2025), and GPT-5-nano (OpenAI, 2025)—score outputs from 180 Chinese topic-comment and 120 English rewriting prompts; the judges are not shown the ground-truth source, and language is coupled with generation task.

The detector probabilities and judge ratings do not move together consistently. In Chinese, SFT raises the RCNN detector’s sarcasm score from 0.780 to 0.895 but lowers the mean LLM sarcasm rating from 3.99 to 3.80; the sarcasm correlation is nearly zero in Chinese (-0.020) but positive in English ($+0.338$). The clearest condition-level reversal concerns human-likeness: SFT raises human-authorship probabilities in both task–language settings while mean perceived human-likeness ratings decline, yielding a pooled correlation of -0.212 . Within-condition correlations are weaker, showing that the pooled negative value is driven mainly by the reversal between conditions.

The sarcasm-detector change is setting-dependent: $+0.115$ in Chinese and -0.124 in English. By contrast, all six language $\times$ judge comparisons give SFT lower mean sarcasm

ratings, five significantly after FDR correction. Detector probabilities can therefore remain useful classification outputs while failing to preserve the ordering or condition-level direction required of style or quality scores.

Agreement among the three judges is low to moderate at the individual-text level. A plain-detector control changes the reported correlations by at most 0.02. The stable human-authorship/human-likeness reversal and the judge-specific sarcasm results therefore motivate external validation before detector probabilities are reused as generation-quality scores or rewards. Full protocols, tests, and illustrative cases are reported in [Appendices C–D](#).

7 Conclusion

LLM-based augmentation requires generated examples to convey sarcasm without turning source cues into shortcuts. Our results expose this risk: compared with a same-architecture single-head control, the plain dual-head model predicts sarcasm for AI text prompted to be non-sarcastic, raising its false-positive rate by 67.0 pp. The direction holds across languages and when the AI test texts are regenerated with Qwen3.5-9B; RCNN and interaction projection show lower rates. Representation interventions show that source-related and other high-variance directions shape how errors are distributed across source groups, providing concrete targets for further causal analysis. Human-authorship probabilities diverge from perceived human-likeness, so detector scores require external, source $\times$ class-stratified validation before use as quality or reward signals.

Limitations

DualTest combines human sarcasm annotations with instruction-conditioned labels for AI texts, so AI-side false positives are defined relative to requested style rather than independent perceived-

sarcasm annotation. Generation failures from deepseek-r1:8b leave 80 human texts without generated counterparts. Some SarcTrain duplicates span validation folds and may affect checkpoint selection; these records do not enter Dual-Test.

The core dual–single comparisons identify the complete joint-training procedure: architecture, sarcasm data, folds, sarcasm-head learning rate, and class weights are shared, while MGTD batches, task scheduling, shared updates, and joint early stopping enter together. Compute-matched controls could further decompose this effect. Cross-architecture results are configuration-level because capacity and optimization differ, and the cost-sensitive loss can affect absolute rates. AuthTrain is generated with deepseek-r1:8b; Qwen3.5-9B replaces only DualTest’s AI side and therefore provides directional test-set replication rather than retraining with another generator.

The fixed-model linear analyses connect source-related structure to behavior but do not isolate a unique causal feature or training-seed distribution. The detector–judge study uses LLM rather than human ratings and couples language with generation task. These boundaries motivate independent style annotation, matched-budget controls, cross-seed mechanism tests, and broader generator and human-evaluation studies.

Ethical considerations

All human-authored texts come from publicly released research datasets: iSarcasm, CSC, ToSarcasm, News Headlines, and the Open Chinese Internet Sarcasm Corpus. Explicit repository licenses were available for iSarcasm and CSC; for sources without an identified redistribution license, we do not redistribute the human-authored text (Appendix A). We collect no new user data. We did not conduct an additional personally identifiable information or offensive-content audit beyond the preparation reflected in the released corpora.

All AI texts were generated locally using publicly released open-weight models (deepseek-r1:8b, Qwen3.5-9B, and the base and QLoRA-adapted Qwen3-8B). They are used only for training and evaluation and are labeled as AI-generated. The study recruits no new human participants, but analyzes human-authored texts from previously released datasets. The automated evaluation transmits each evaluation

input, generated output, and rating instructions to the DeepSeek-V3, Gemini-2.5-Flash-Lite, and GPT-5-nano APIs; it does not transmit dataset metadata or ground-truth source labels.

The detectors are developed for research diagnosis. Their outputs should not be used as evidence of authorship or misconduct at the individual-text level without human review and domain-specific validation, nor deployed for content moderation or automated sanctions without evaluating out-of-distribution behavior. The class-conditional failure diagnosed here is an example of systematic subgroup error affecting a particular type of generated content.

AI-assisted tools were used for grammar correction, wording suggestions, and formatting. They did not generate the reported experimental outputs or statistics; the authors determined the study design, analysis, and conclusions and take full responsibility for the work.

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A Detailed Dataset Statistics and Quality Checks

Corpus	Lang.	Inst.	Use
iSarcasm	English	3,467	Sarc. train + SFT
(SemEval-2022 T6)			
CSC	English	6,878	Sarc. train
ToSarcasm	zh 3,892 / en 6	3,898	Sarc. train + SFT
News Headlines v2	English	28,619	Sarc. train
Open Chinese Internet Sarcasm Corpus	Chinese	2,000	Sarc. train

Table 4: Input corpora. Genre / annotation: iSarcasm — tweets, speaker self-annotation; CSC — conversational responses; ToSarcasm — topic comments, third-party annotation; News Headlines — distant supervision (The Onion / HuffPost); Open Chinese Corpus — Weibo, third-party annotation.

A.1 Corpus details. The pooled corpus collection contains 44,862 instances in total (86.9% English / 13.1% Chinese). Chinese–English 1:1 balance in SarcTrain is achieved through bucketed

sampling. The two English buckets and the Chinese non-sarcastic bucket contain more than 2,500 available instances and are randomly downsampled within each bucket. The Chinese sarcastic bucket contains 2,396 source records representing 2,394 unique texts; 104 records are added by oversampling to reach 2,500. Including source-level duplicate text, the final bucket’s text-duplication rate is 4.24% (A.3).

Licensing and release scope. The iSarcasm and CSC repositories state MIT licenses. We did not identify explicit redistribution licenses in the releases we accessed for ToSarcasm, News Headlines v2, or the Open Chinese Internet Sarcasm Corpus. We therefore do not redistribute human-authored text from those sources. Planned research artifacts are limited to eligible AI-generated text, labels and metadata, and reconstruction scripts that require users to obtain the human-authored corpora from their original sources.

Lang.	Source	Sarc.	Non-sarc.
English	Human	200	1,200
English	AI	192	1,174
Chinese	Human	623	349
Chinese	AI	587	339

Table 5: Language $\times$ source $\times$ sarcasm distribution of DualTest. Human labels are dataset annotations; AI labels record prompted conditions.

A.2 Three-dimensional distribution of DualTest.

A.3 Data quality checks. Automated checks confirm language and label balance across the five SarcTrain folds. AuthTrain contains 9,996 complete human–AI pairs and three unmatched human instances. Record-level matching confirms that neither SarcTrain nor AuthTrain shares records with the human or AI side of DualTest; no train–test leakage is present.

SarcTrain’s text-duplication rate is 0.34% for English / 2.16% for Chinese. Chinese duplication is concentrated in the sarcastic bucket (4.24%; non-sarcastic bucket: 0.08%) and results from the oversampling used to fill that bucket (A.1). Because oversampling precedes fold assignment, 86 of the 104 added records have a source counterpart in another fold. Under the raw-text counting convention used for these duplication rates, 116 repeated-text groups involving 241 records are present when

oversampled copies are included, and 98 groups span folds. Excluding the added copies and normalizing whitespace and case leaves 19 naturally duplicated groups involving 46 records, of which 16 span folds. These relationships are relevant to fold-based checkpoint selection but do not place the affected training or validation records into DualTest. Post-processing also checks for the explicit chain-of-thought markers and reasoning prefaces described in A.4 (Wei et al., 2022).

A.4 AI generation protocol. The AI texts in AuthTrain and the two DualTest versions are produced by locally deployed generation models through the Ollama API. The model tagged deepseek-r1:8b generates AuthTrain and the DeepSeek version of DualTest’s AI side; Qwen3.5-9B under tag qwen3.5:9b generates the alternate AI side used for cross-generator replication. Both use temperature = 0.8, top- p = 0.95 (Holtzman et al., 2020), and a generation limit of 500 tokens. Generation instructions use four templates defined by language $\times$ style, with style determined by the seed instance’s sarcasm label. The templates constrain output style and length (20–80 Chinese characters; 10–50 English words). This constraint is intended to reduce, but does not eliminate, source differences in length; Appendix E separately repeats the representation correlation analysis within length tertiles. The complete Chinese sarcastic template used in the experiment is reproduced verbatim below:

【任务】生成一条讽刺风格的评论
 【主题】{seed}
 【要求】
 - 直接给出简短的评论，不要解释思路或分析
 - 风格要讽刺、幽默、夸张
 - 表达与主题相关的观点即可，不必追求严谨
 - 20–80 字为佳
 【评论】

Its pinyin transliteration is:

[rènwù] shēngchéng yī tiáo fěngcǐ
 fěnggé de pínglùn
 [zhǔtí] {seed}
 [yāoqiú]
 - zhíjiē gěichū jiǎnduǎn de pínglùn,
 bù yào jiěshì sīlù huò fēnxī
 - fěnggé yào fěngcǐ, yōumò, kuāzhāng
 - biǎodá yǔ zhǔtí xiāngguān de guāndiǎn
 jíkě, bù bī zhuīqiú yánjīn
 - 20–80 zì wéi jiā
 [pínglùn]

Its English translation is:

```
[Task] Generate a sarcastic-style
comment
[Topic] {seed}
[Requirements]
- Directly provide a short comment;
do not explain your reasoning or
analysis
- The style should be sarcastic,
humorous, and exaggerated
- It is sufficient to express a
viewpoint related to the topic;
strict rigor is not required
- Preferably 20--80 Chinese
characters
[Comment]
```

The Chinese non-sarcastic template changes only the style line to “The style should be normal and natural.” The English templates follow the same structure, using either “sarcastic, humorous, exaggerated” or “normal, natural,” with a length limit of 10–50 words.

Post-processing removes `<think>...</think>` segments, reasoning prefaces such as “好的” (*hǎo de* ‘okay’), “嗯” (*ní* ‘hmm’), “Okay,” “Let me,” and “First,” and Markdown markup. Outputs that remain overlength or cyclic after repeated generation attempts are discarded. AuthTrain retains 9,996 AI counterparts from 9,999 seeds; DualTest retains 2,292 counterparts from 2,372 seeds.

The Qwen3.5-9B replication has no generation failures, yielding 1,550 AI $\times$ non-sarcastic instances rather than 1,513. Generator-specific rates therefore use each retained full output set rather than an identical seed intersection. For this run, Ollama thinking mode is disabled (`think = false`) to prevent the reasoning process from exhausting the output budget. The same post-processing removes any inline chain-of-thought from the `deepseek-r1:8b` outputs.

A.5 SFT training and generation-evaluation set.

Eligibility filtering retains 5,257 positive-sarcasm records suitable for generation tasks (2,972 English; 2,285 Chinese). A seed-42 shuffle and 40% sampling select 2,103 records (1,214 English; 889 Chinese); language balancing then adds 325 oversampled Chinese records, producing GenTrain (2,428; 1,214 per language). By prompt type, it comprises 374 rewriting, 839 response, and 1,215 comment instances. Generation evaluation uses an independent set of 300 prompts (180 Chinese and 120 English): the Chinese task is to generate a sarcastic comment on a given topic, and the

English task is to rewrite a normal sentence as a sarcastic one. The detector scoring and three source-blinded LLM evaluations in §6 are both conducted on this set.

B Hyperparameters and Training Details

Group	Enc.	Inter.	Sarc.	MGTD
Plain / backbone dual	5e-7	—	5e-6	1e-5
RCNN dual	5e-7	5e-6	5e-6	5e-6
Interaction-projection dual	5e-7	5e-6	5e-6	5e-6
Plain single-task	5e-7	—	5e-6	1e-5
RCNN single-task	5e-7	5e-6	5e-6	1e-5

Table 6: Discriminative learning rates for the evaluated dual- and single-task configurations.

B.1 Discriminative learning rates. The sarcasm-head learning rate is fixed at 5e-6 across configurations. The MGTD-head rate is 1e-5 for the plain dual and single-task models, 5e-6 for the two intermediate dual-head variants, and 1e-5 for the RCNN MGTD single-task control. Thus, the core sarcasm dual–single comparisons retain the same sarcasm-specific rates within each architecture, whereas the RCNN MGTD comparison and cross-configuration results can also reflect optimization settings. Early stopping uses the normalized joint validation loss of Equation 2 for dual-head models and the task’s own validation loss for the single-task controls, with the same patience of five epochs in both cases.

B.2 Shared training hyperparameters. Training uses AdamW with default PyTorch betas (0.9, 0.999) and $\epsilon = 10^{-8}$, weight decay 0.01, and parameter-group learning rates in B.1. A LambdaLR schedule linearly warms up for 200 steps and then keeps the learning rate constant. Training uses at most 10 epochs, early-stopping patience 5, and gradient clipping 1.0. The physical batch size is 4 with gradient accumulation 4, giving an effective batch size of 16. The MGTD:sarcasm sampling ratio is 2:1, and sarcasm class weights are [1.0, 2.5]. Both choices were specified before testing. With SarcTrain class-balanced, the class weights implement a cost-sensitive objective that assigns a greater penalty to errors on sarcastic examples. The normalized joint validation loss is $0.5 \times [\text{sarcasm validation loss} / \text{first-epoch value}] + 0.5 \times [\text{MGTD validation loss} / \text{first-epoch value}]$. The single-head controls traverse the full sarcasm

training fold each epoch, whereas the dual-head models draw sarcasm batches under the 2:1 task-mixing schedule.

Detector inputs use a maximum sequence length of 512 tokens. In the RCNN module, a width- k convolution uses padding $\lfloor k/2 \rfloor$ followed by ReLU; global max pooling is applied directly over the full convolution output without an additional padding mask.

All fold-based runs start from seed 42 plus the fold index and share the same fold assignments and nominal seed scheme across architectures and ablations. MGTD data use a fixed 90/10 holdout in every fold. For each final-model configuration, the fixed epoch count is the mode of its five best epoch counts plus one, capped at 10 epochs; the resulting counts range from 3 to 10 across configurations. The final model uses all applicable training data, seed +1000, no validation set, and no early stopping. Main tables report five-run means and standard deviations rather than final-model point estimates.

Model sizes, software, and hardware environment. The encoders have approximately 178M (mBERT), 278M (XLM-RoBERTa-base), 184M (DeBERTa-v3-base), and 435M (DeBERTa-v3-large) parameters; the SFT generator Qwen3-8B has about 8B parameters, of which QLoRA updates only low-rank adapters. Detector training and inference are performed on a single NVIDIA GeForce RTX 3090 (24 GB); the five-fold training of the backbone comparison, architecture, and ablation variants together with the final model amounts to on the order of tens of GPU-hours, with a single detector run taking from a few minutes to a few hours and SFT about two hours. The software environment is PyTorch 2.9.0 (CUDA 12.8), transformers 4.57.1, and peft 0.18.0. AI texts are generated locally through Ollama (Appendix A.4); blinded LLM evaluation uses API calls made in May 2026 (Appendix C.3).

B.3 SFT (QLoRA). We fine-tune Qwen3-8B using QLoRA (Detmers et al., 2023), with 4-bit loading and bfloat16 computation. The LoRA adapters (Hu et al., 2022a) use rank 64, alpha 16, dropout 0, and target q/k/v/o_proj plus gate/up/down_proj. The maximum sequence length is 2,048, with gradient checkpointing enabled. Training uses learning rate 5e-5, three epochs, batch size 1, gradient accumulation 8, and warmup ratio 0.05. After language balancing, a seed-42 split holds out 15% of

GenTrain for validation; validation loss selects the checkpoint.

C Complete Experimental Results

C.1 Macro-averaged F1 (supplement to Table 1). RCNN macro-averaged F1, under the same five-run protocol and calculated from each run’s confusion matrix, is 0.6678 ± 0.036 for sarcasm and 0.9298 ± 0.005 for MGTD. Sarcasm macro-F1 is higher than its positive-class F1 (0.6309) because the majority class has a higher F1. These values supplement rather than rerank the full positive-class comparison in §5.1.

C.2 Complete four-quadrant results for the MGTD head (including single-head controls). The reporting convention matches Table 3 (recall for human subsets; false-positive rate for AI subsets).

Model	H×s	H×n	AI×s	AI×n
Plain dual	91.4	91.5	3.6	6.1
Inter.-proj. dual	85.4	85.4	1.8	4.1
RCNN dual	94.8	93.2	5.5	6.6
Plain single	90.4	86.6	4.1	4.2
RCNN single	90.5	92.4	3.6	5.7

Table 7: MGTD-head four-quadrant results (%): recall for human (H) subsets, false-positive rate for AI subsets. s = sarcastic, n = non-sarcastic.

The single-head MGTD controls remain within single-digit percentage points of the dual-head models in every quadrant. No analogous extreme class-conditional error appears on the MGTD side, consistent with §5.3.

No comparable subgroup increase appears on the qwen3.5:9b replication set: AI × non-sarcastic false-positive rates are 0.9%, 1.3%, and 0.5% for plain, interaction projection, and RCNN; AI × sarcastic rates are at most 0.4%; and overall MGTD F1 scores are 0.952, 0.917, and 0.966.

C.3 Blinded-evaluation protocol, inter-judge agreement, and group-level tests (§6). Each sample is sent separately to the three judge APIs with one prompt (May 2026; temperature = 0.3; GPT-5-nano uses its default decoding because it does not support this parameter). The prompt contains only the input and evaluated text, not the ground-truth source. It asks for source inference followed by 1–5 ratings of sarcasm and human-likeness. Base and SFT outputs use the same prompt and are evaluated separately.

Pairwise instance-level agreement is moderate to low. Spearman ρ ranges from 0.20 to 0.38 for sarcasm (all $p < 10^{-6}$) and from 0.03 to 0.23 for human-likeness (one pair is not significant). The conclusion in §6 therefore rests on consistent condition-level directions, not agreement in instance rankings.

Wilcoxon signed-rank tests with Benjamini-Hochberg FDR correction (Benjamini and Hochberg, 1995) show that SFT receives lower sarcasm ratings than the base model in five of six language $\times$ judge groups. The pooled correlations in Table 8 use the three-judge mean as a descriptive aggregate; because inter-judge agreement is limited, the judge-specific results remain necessary. Four unparseable ratings occur across four instances, each of which retains two valid judge ratings; the aggregate uses the mean of available ratings and therefore retains all 600 outputs.

Subset	n	Sarc. ρ	MGTD ρ
Overall	600	+0.136*	−0.212*
Chinese	360	−0.020	−0.218*
English	240	+0.338*	−0.270*

Table 8: Spearman correlations between detector probabilities and LLM-judge ratings (RCNN detector, $n = 600$ pooled over base and SFT). * significant; other entries not significant.

Judge / lang.	n	SFT	Base	adj. p
DeepSeek-V3 / en	120	3.292	4.025	<.0001
DeepSeek-V3 / zh	180	3.878	4.044	.0008
Gemini-2.5-FL / en	120	3.750	3.992	.0008
Gemini-2.5-FL / zh	180	3.994	4.028	.2008*
GPT-5-nano / en	119	3.681	4.084	<.0001
GPT-5-nano / zh	180	3.517	3.911	<.0001

Table 9: Group-level tests for blinded sarcasm judgments (Wilcoxon + BH-FDR). * not significant. GPT-5-nano lacks one English base rating (C.3: $n = 119$) and three Chinese reference ratings (affecting C.4, not the C.3 Chinese row); tests use pairwise deletion.

C.4 Paired tests between SFT and human references (supplementary). This test is performed on the Chinese subset of the generation-evaluation set ($n = 180$; GPT-5-nano is missing three instances, leaving 177). These comparisons provide judge- and dimension-specific calibration context: for DeepSeek-V3, SFT is significantly

below the reference on human-likeness (-0.161 , $p = 0.0025$), whereas for GPT-5-nano, SFT is significantly above the reference on sarcasm ($+0.186$, $p = 0.0114$). The remaining four comparisons are statistically indistinguishable at the reported threshold, showing that proximity to the human reference depends on the judge and evaluated dimension.

C.5 Plain-detector control. With the plain detector, Chinese sarcasm changes from 0.765 to 0.763 (-0.001), English sarcasm from 0.843 to 0.659 (-0.184), and the MGTD score rises by $+0.760$ in Chinese and $+0.705$ in English. The two detectors therefore agree on the rise in MGTD scores, while the sarcasm direction depends on the detector.

Correlations are $+0.142$ for sarcasm and -0.214 for MGTD overall; -0.037 (not significant) and -0.209 in Chinese; and $+0.342$ and -0.283 in English. Each differs from the RCNN-detector value by at most 0.02. Negative human-authorship-probability versus human-likeness associations are similar across the two detectors, whereas the condition-level sarcasm-score change is architecture-dependent, especially in Chinese.

Within-condition correlations (RCNN detector). For MGTD, the base condition has $\rho = +0.03$ ($n = 300$, not significant; Chinese $+0.10$, English -0.09). The SFT condition has $\rho = -0.12$ ($p = 0.039$; Chinese -0.19 , $p = 0.009$; English -0.12 , not significant). The pooled value of -0.212 is driven mainly by the opposing condition-level changes: SFT raises the detector score by about 0.67 while lowering the mean blinded rating. There is no positive detector-judge alignment on this dimension within conditions.

For sarcasm, within-condition correlations are $+0.05$ for Chinese base (not significant), $+0.15$ for Chinese SFT ($p = 0.048$), $+0.41$ for English base, and $+0.37$ for English SFT (both English values significant). Chinese alignment is therefore at most weakly positive, consistent with its near-zero pooled result.

C.6 Detector scores before and after SFT across the two task-language settings. Scores from the RCNN detector for base and SFT outputs (0–1):

Sarcasm scores move in opposite directions in the Chinese comment-generation and English rewriting settings: $+0.115$ and -0.124 , respectively. GenTrain itself is balanced at 1,214 in-

Lang.	Cond.	Sarc.	MGTD
Chinese	Base → SFT	0.780 → 0.895	0.289 → 0.969
English	Base → SFT	0.854 → 0.730	0.198 → 0.864

Table 10: Detector scores before/after SFT (RCNN detector).

stances per language; the evaluation settings instead differ in task, prompt format, source data, and sample size, all of which may contribute to the contrast. MGTD scores rise sharply in both settings, so the detector assigns SFT outputs higher probabilities of human authorship.

Mean LLM judgments for sarcasm / human-likeness are 3.77/4.17 for human references, 4.01/4.24 for the base model, and 3.71/4.08 for SFT. These directions conflict with the detector’s MGTD score, as analyzed in §6.

C.7 Conditional test-instance uncertainty: confidence intervals, paired tests, and subset F1.

Conditional on each fixed final model, we quantify test-instance uncertainty with Wilson 95% intervals for quadrant rates and 95% percentile intervals from 10,000 instance-level bootstrap samples for F1. These intervals do not include uncertainty from training, generator sampling, corpus construction, or human–AI pairing. They complement the between-run standard deviations in Table 2 and Table 1. All values use the final model for each architecture and follow the reporting convention of Table 3.

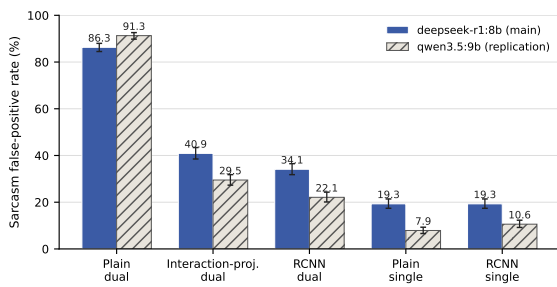


Figure 2: Sarcasm false-positive rates on the AI × non-sarcastic subset for five detectors and two generators. Error bars are Wilson 95% intervals conditional on each fixed model.

The fixed final-model configurations show the same directional dual–single gap on the Qwen3.5-generated AI texts, although its magnitude differs.

Paired McNemar tests use predictions on the same DeepSeek-generated AI × non-sarcastic texts. Discordant pairs are 1,021 / 7 for plain dual

Model	DeepSeek	Qwen3.5
Plain dual	86.3 [84.5, 88.0]	91.3 [89.8, 92.6]
Inter.-proj. dual	40.9 [38.5, 43.4]	29.5 [27.3, 31.8]
RCNN dual	34.1 [31.8, 36.5]	22.1 [20.1, 24.3]
Plain single	19.3 [17.4, 21.4]	7.9 [6.6, 9.3]
RCNN single	19.3 [17.4, 21.4]	10.6 [9.2, 12.3]

Table 11: Wilson 95% intervals for the sarcasm false-positive rate on the AI × non-sarcastic subset (two generators; %).

vs. plain single, 239 / 15 for RCNN dual vs. RCNN single, 805 / 15 for plain dual vs. RCNN dual, and 203 / 100 for interaction projection vs. RCNN dual (all $p < .001$). The discordant counts and absolute rate differences carry the substantive effect size.

Model	Lang.	n	Dual pos.; rate [CI]	Single pos.; rate [CI]	Δ pp
Plain	en	1,174	987; 84.1 [81.9, 86.1]	242; 20.6 [18.4, 23.0]	+63.5
Plain	zh	339	319; 94.1 [91.1, 96.2]	50; 14.7 [11.4, 18.9]	+79.4
RCNN	en	1,174	445; 37.9 [35.2, 40.7]	247; 21.0 [18.8, 23.5]	+16.9
RCNN	zh	339	71; 20.9 [17.0, 25.6]	45; 13.3 [10.1, 17.3]	+7.7

Table 12: Language-stratified sarcasm false-positive rates on AI texts prompted to be non-sarcastic (%; Wilson 95% intervals). Exact paired McNemar tests give $p < .001$ for both plain comparisons and RCNN English, and $p = 2.4 \times 10^{-5}$ for RCNN Chinese.

Both same-architecture comparisons therefore have the same direction in Chinese and English. Language remains coupled with corpus, genre, and annotation setting, so these results provide stratified replication across the two task–language settings rather than an isolated language effect.

The final RCNN dual-head model’s MGTD F1 is 0.938 [0.931, 0.945] on the full set, 0.961 [0.952, 0.969] in Chinese, and 0.923 [0.912, 0.933] in English. Because the human subset contains no AI-labeled negatives, a source-class F1 is not separately reported for this subset; Table 7 reports human recall directly.

Chinese sarcasm F1 is higher than English sarcasm F1 for the RCNN model (0.819 vs. 0.429). The contrast is confounded by prevalence, genre, annotation procedure, and language: sarcasm prevalence is 63.8% in Chinese but 14.2% in English, and the English human side is dominated by author-labeled intended-sarcasm tweets from iSarcasmEval whereas the Chinese side uses topic-comment-style ToSarcasm. The positive-class F1 values are therefore not directly comparable as a pure language contrast.

The separately retrained final RCNN model’s overall sarcasm F1 is 0.663, compared with 0.629

Model	Full set	Chinese	English	Human subset	Human-zh	Human-en
Plain dual-head	0.556	0.746	0.312	0.586	0.710	0.374
Interaction-projection dual-head	0.643	0.781	0.420	0.593	0.739	0.366
RCNN dual-head	0.663 [0.646, .680]	0.819 [0.802, .836]	0.429 [0.399, .458]	0.629 [0.605, .653]	0.774 [0.748, .798]	0.399 [0.355, .440]
Plain single-head	0.661	0.750	0.514	0.612	0.719	0.434
RCNN single-head	0.664	0.758	0.511	0.603	0.709	0.434

Table 13: Per-language and human-subset sarcasm F1 (final models; bootstrap 95% intervals included for RCNN).

on human texts alone, because the overall score also includes AI texts. The five-run mean in Table 1 is a different estimate. Within the human subset, dual-single differences are below 0.03 and vary in direction. The visible negative-transfer loss is therefore concentrated in the AI-side quadrants, consistent with §5.3.

False-positive rates use a threshold of 0.5. For the plain dual-head model on AI $\times$ non-sarcastic text, the median sarcasm probability is 0.767 with an interquartile range of 0.635–0.839. The distribution therefore shifts across the threshold rather than placing only a few cases near it. Single-head medians are 0.204 for the plain model and 0.069 for RCNN.

AUC values within AI texts are similar across models (0.82–0.88). Joint training thus raises sarcasm scores while largely preserving their internal ordering. This pattern is consistent with a source-related component being added to the sarcasm signal.

D Illustrative Cases of Detector–LLM Score Misalignment

Illustrative cases were manually selected after inspecting the paired table of RCNN-detector scores and ratings from the three LLMs to show large disagreements; no formal ranking or top- k selection was implemented. Detector scores range from 0 to 1, and LLM scores are the three-evaluator mean on a 1–5 scale; long texts are truncated. Chinese examples retain the original model input and provide a pinyin transliteration and an English translation in brackets.

The three groups show, respectively, high detector human-authorship scores with low LLM human-likeness ratings, high detector sarcasm scores with low LLM sarcasm ratings, and the reverse sarcasm disagreement. In the first human-likeness case, AI-generated text with obvious chain-of-thought traces is nevertheless assigned human-authorship probability 1.00 by the MGTD head, corresponding to the score disagreement discussed in §6.

L	S	$P_{\text{hum}}/\text{LLM}$	Text (excerpt)
zh	SFT	1.00 / 2.67	“好，我需要针对三星回怼华为的新闻生成一条讽刺评论。首先……” [hǎo, wǒ xūyào zhēnduì Sānxīng huíduì Huáwéi de xīnwén shēngchéng yī tiáo fěngcì pínglùn. shǒuxiān...] [“Okay, I need to generate a sarcastic comment on the news of Samsung hitting back at Huawei. First...”] (chain-of-thought leakage)
zh	SFT	1.00 / 2.33	“跑分竞赛：参数焦虑：跑分焦虑：水货：哈哈哈” [pǎofēn jìngsài: cānshù jiāolǜ: pǎofēn jiāolǜ: shuǐhuò: hāhāhā] [“Benchmark race: spec anxiety: benchmark anxiety: grey-market goods: hahaha”] (degenerate output)
en	SFT	1.00 / 2.67	“Same as me, I’m not sure” (vacuous short sentence)

Table 14: MGTD-score misalignment cases. $P_{\text{hum}} = P(\text{human})$.

L	S	$P_{\text{sar}}/\text{LLM}$	Text (excerpt)
zh	SFT	1.00 / 2.67	“三星：我才是第一！你看看我这摄像头总分，109+96，112+89，我赢了！” [Sānxīng: wǒ cái shì dì-yī! nǐ kànkān wǒ zhè shèxiàngtóu zǒngfēn, 109+96, 112+89, wǒ yíng le!] [“Samsung: I’m the real number one! Look at my camera total scores, 109+96, 112+89, I win!”] (numbers repeated without sarcastic transformation)
zh	SFT	0.99 / 2.67	“恭喜恭喜，美国牛牛牛，继续加油” [gōngxǐ gōngxǐ, Měiguó niú niú niú, jìxù jiāyóu] [“Congratulations, congratulations, America is awesome awesome awesome, keep it up”] (formulaic mock-praise)
en	SFT	0.93 / 2.67	“2 hours left to tip it over, I’m going to the toilet”

Table 15: Detector-high / LLM-judge-low sarcasm cases. $P_{\text{sar}} = P(\text{sarcastic})$.

L	S	$P_{\text{sar}}/\text{LLM}$	Text (excerpt)
zh	Base	0.14 / 4.00	“特朗普承诺 120 亿补贴像撒大饼，农民收到的却是缩水成硬币的承诺……” [Tèlángpǔ chéngnuò 120 yì bǔtiē xiàng sā dàbǐng, nóngmín shōudào de què shì suōshuǐ chéng yìngbì de chéngnuò...] [“Trump promised a 12-billion subsidy as if tossing out big pancakes, yet what farmers receive are promises shrunk into coins...”] (metaphorical irony)
zh	Base	0.24 / 4.00	“特斯拉：价格跳水后又跳水，消费者：钱包跳水后又跳水。” [Tèsīlā: jiàgé tiàoshuǐ hòu yòu tiàoshuǐ, xiāofèizhě: qiánbāo tiàoshuǐ hòu yòu tiàoshuǐ.] [“Tesla: prices dive and then dive again; consumers: wallets dive and then dive again.”] (parallel-structure irony)
en	Base	0.12 / 4.00	“Thank you so much for this *beloved* review, Mandy! I’m *overjoyed* you enjoyed it” (asterisk emphasis)

Table 16: Detector-low / LLM-judge-high sarcasm cases. $P_{\text{sar}} = P(\text{sarcastic})$.

Together, these selected cases motivate the hypothesis that surface markers such as mock-praise formulas, exclamation marks, and colloquial short sentences contribute to some detector decisions, while semantic reversal contributes to some LLM-judge ratings. They define candidate features for systematic prevalence and feature-level testing and provide qualitative examples of the near-zero Chinese sarcasm correlation in §6.

E Representation-Level Analysis of Source-Conditioned Error Allocation

This appendix records the complete protocol, data, and self-checks for the representation-level tests in §5.3.2. The interpretation of conclusions follows §5.3.2.

E.1 Setup. All analyses freeze the plain dual-head model that produces the 86.3% main-text result. We use split-sample evaluation: one half estimates the direction and the other computes metrics. The appendix baseline is therefore 84.8% on the evaluation half rather than 86.3% on the full set.

Implementation checks pass. Manually computed logits match model outputs with maximum error 1.4×10^{-6} , and the erasure-projection residual is about 10^{-6} . LEACE satisfies its defining property: after erasure the fitting-half class-mean difference has relative residual 7×10^{-6} , and an in-sample linear probe is at chance (AUC 0.514). The source-probe AUC is 0.986 with true labels and 0.512 under source-label permutation.

E.2 Protocol. The linear probe is logistic regression with five-fold cross-validation. A permuted-label control retrains the same probe after shuffling source labels (Hewitt and Liang, 2019), and cor-

relation analyses are repeated within text-length tertiles. For alignment, Σ is the Ledoit–Wolf shrinkage covariance estimator (Ledoit and Wolf, 2004) fitted to all pooled DualTest representations, with eigenvalues clipped below 10^{-8} ; w denotes a task-head weight difference or probe direction. Whitened alignment is $\cos(\Sigma^{1/2}w_1, \Sigma^{1/2}w_2)$.

Erasure tests cover one direction, INLP subspaces ($k = 1\text{--}16$; Ravfogel et al., 2020), and LEACE (Belrose et al., 2023). Controls use ten sampled random whitened directions and directions estimated from permuted labels. We construct source-direction-orthogonal PC1 (srcORTH-PC1) by projecting the fitting-half representations onto the orthogonal complement of the estimated linear source direction and then extracting the maximum-variance direction in that projected space. Erasure removes each representation’s scalar projection onto this direction. This operation leaves measured linear source decodability unchanged, although orthogonality to one estimated direction does not imply independence from all multidimensional or nonlinear source information.

E.3 Diagnostic evidence. Source is highly linearly decodable: AUC is 0.986 overall, 0.990 on sarcastic texts, and 0.985 on non-sarcastic texts. The permuted-label control gives 0.512. The single-head model also gives 0.974, establishing source decodability as a background property shared by both models. On AI $\times$ non-sarcastic texts, the MGTD-head AI logit and sarcasm-head logit correlate at $\rho = 0.519$; correlations by text-length tertile are 0.44, 0.56, and 0.63.

The sarcasm and source directions are nearly orthogonal in the original geometry ($\cos \approx -0.03$) but moderately aligned after whitening. The absolute cosine is 0.41 using the probe direction and 0.49 using the MGTD-head direction. These results provide correlational evidence consistent with the shortcut hypothesis and motivate the intervention tests below.

E.4 Intervention evidence. $\text{gap} = \text{AI} \times \text{non-sarcastic false-positive rate} - \text{human} \times \text{non-sarcastic false-positive rate}$; source AUC is the decoding ability of a probe retrained after erasure.

Single-direction and low-dimensional INLP erasure produce little change. LEACE reduces measured linear source decodability and the source-specific gap from 52.0 to 19.1; the ten sampled random-direction controls show no comparable

Erasure scheme	AI×n FP	H×n FP	Gap	AUC
None (baseline)	84.8	32.8	52.0	0.982
Single dir. (head / probe / random)	84.8–86.2	—	—	≈0.98
INLP ($k = 16$)	89.8	—	—	0.974
LEACE (linear)	68.7	49.5	19.1	0.589
Random	84.9	32.7	—	0.982
whitened ×10				
srcORTH-PC1	66.7	54.1	12.6	0.982

Table 17: Intervention evidence. H×n = human × non-sarcastic; AI×n = AI × non-sarcastic; FP in %.

gap change.

The orthogonal high-variance control changes the interpretation. It leaves measured linear source decodability intact but changes the AI-side rate to 66.7 and the source gap to 12.6, an effect similar to LEACE. Together with the random-direction controls, this shows prediction sensitivity to a non-random high-variance direction after removing its projection onto the estimated source direction and makes the direction’s semantic content and causal role concrete targets for further analysis.

Under LEACE, the unweighted mean false-positive rate across the two non-sarcastic source groups is almost unchanged ($58.8 \rightarrow 59.1$). This result shows that the affected directions shape how non-sarcastic errors are allocated between AI and human texts.